



Latency Budget

Author : Bouzara Zakaria

Difficulty Level : Easy

Latency Budget

A user clicks a button. What they do not see is their request silently threading through a chain of microservices: authentication, business logic, database, cache. Each service runs multiple replicas at any given moment, and each replica responds at a slightly different speed depending on its current load. The system has an SLA guarantee: if the total round-trip exceeds the latency budget, the request is dropped and the user sees an error.

Your job is to figure out how many valid ways a request can be routed through the service chain without breaching the budget.

Task: Given a sequential chain of N services where each service has a list of replica latencies, count the number of ways to pick exactly one replica per service such that the sum of chosen latencies does not exceed the budget B .

Input

$$\begin{array}{c} N \quad B \\ k_0 \quad l_{(0,0)} \quad l_{(0,1)} \quad \dots \quad l_{(0,k_0-1)} \\ k_1 \quad l_{(1,0)} \quad l_{(1,1)} \quad \dots \quad l_{(1,k_1-1)} \end{array}$$

- **Line 1:** Two integers N (number of services) and B (latency budget).
- **Next N lines:** Each line starts with k_i (number of replicas for service i), followed by k_i space-separated integers: the latency of each replica.

Output

Print a single integer: the number of valid replica combinations whose total latency is at most B .

Constraints

- $1 \leq N \leq 10$
- $1 \leq k_i \leq 10$
- $1 \leq l_{i,j} \leq 100$
- $1 \leq B \leq 1000$
- Time limit: **1 second**
- Memory limit: **256 MB**

Sample Input 1

3 40
3 10 20 15
2 5 30
3 8 12 25

Sample Output 1

7

Explanation

All valid combinations summing to at most 40:

Path	Sum
10 + 5 + 8	23
10 + 5 + 12	27
10 + 5 + 25	40
15 + 5 + 8	28
15 + 5 + 12	32
20 + 5 + 8	33
20 + 5 + 12	37

Sample Input 2

2 4
2 1 2
2 1 2

Sample Output 2

4

Explanation

All four combinations are valid: $(1,1)=2$, $(1,2)=3$, $(2,1)=3$, $(2,2)=4$.